

A **polynomial function** in standard form is

$$f(x) = a_nx^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0, \qquad a_n \neq 0,$$

where the coefficients are real numbers and the exponents are nonnegative integers.

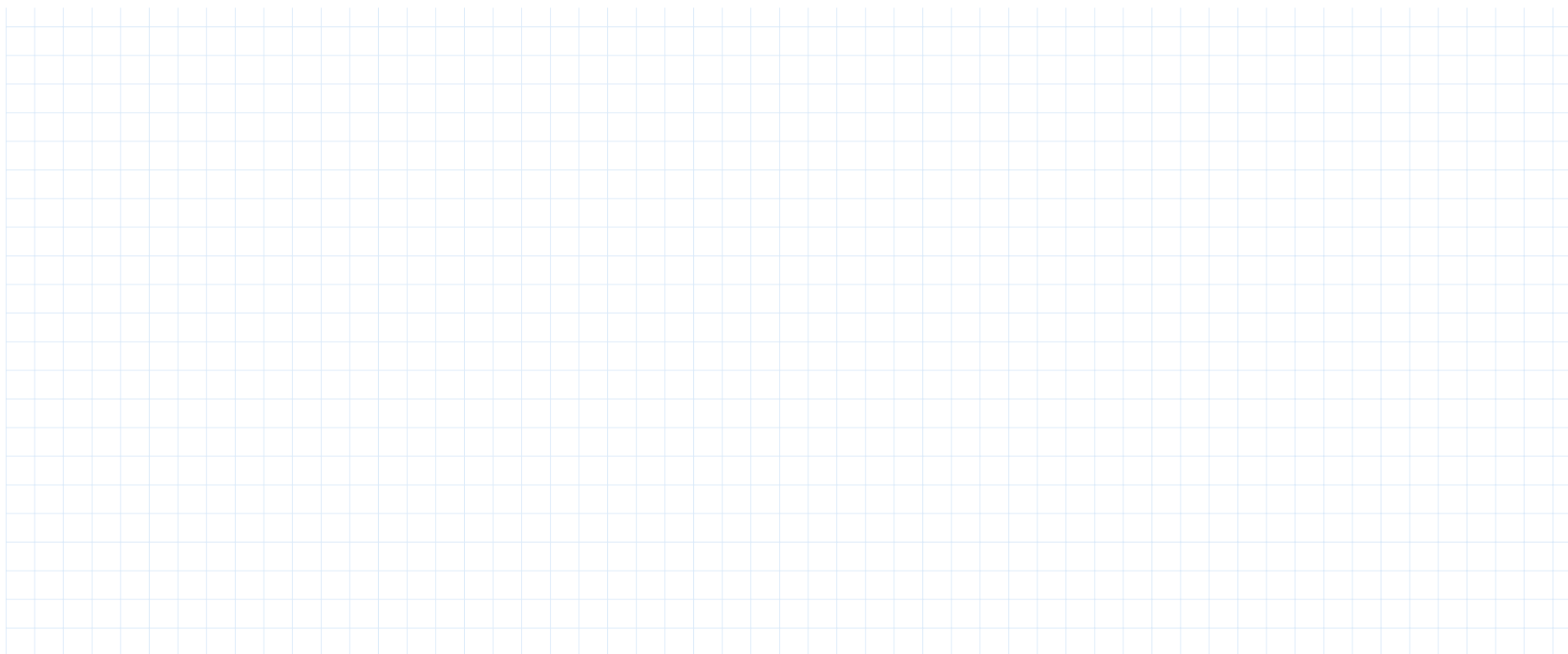
Degree	n , the largest exponent with a nonzero coefficient
Leading term	a_nx^n
Leading coefficient	a_n
Constant term	$a_0 = f(0)$

Every polynomial has domain $(-\infty, \infty)$ and a smooth, continuous graph.
A nonzero constant polynomial has degree 0. The zero polynomial has no defined degree.

Recognize a Polynomial

Example. Which functions are polynomials? Explain any exclusions. For each polynomial, write standard form and identify its degree, leading coefficient, and constant term.

(a) $p(x) = 7 - 3x + 2x^4$ (b) $q(x) = \sqrt{x} + 1$ (c) $r(x) = \frac{x^2 - 4}{x - 2}$ (d) $s(x) = 8$



Power Functions

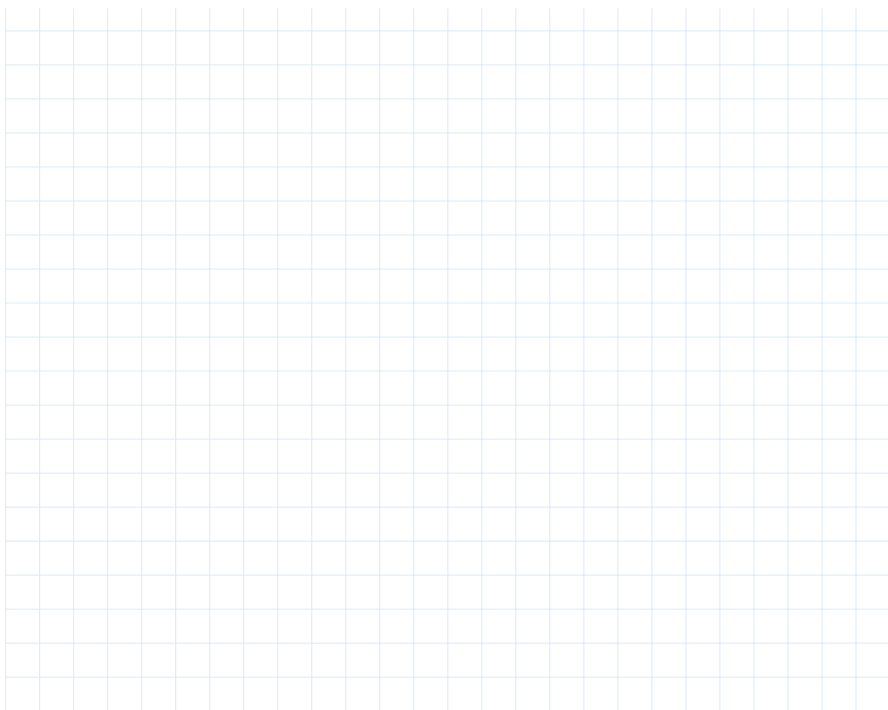
For a positive integer n , $f(x) = ax^n$ with $a \neq 0$ is a polynomial power function.

Even n : symmetry about the y -axis.

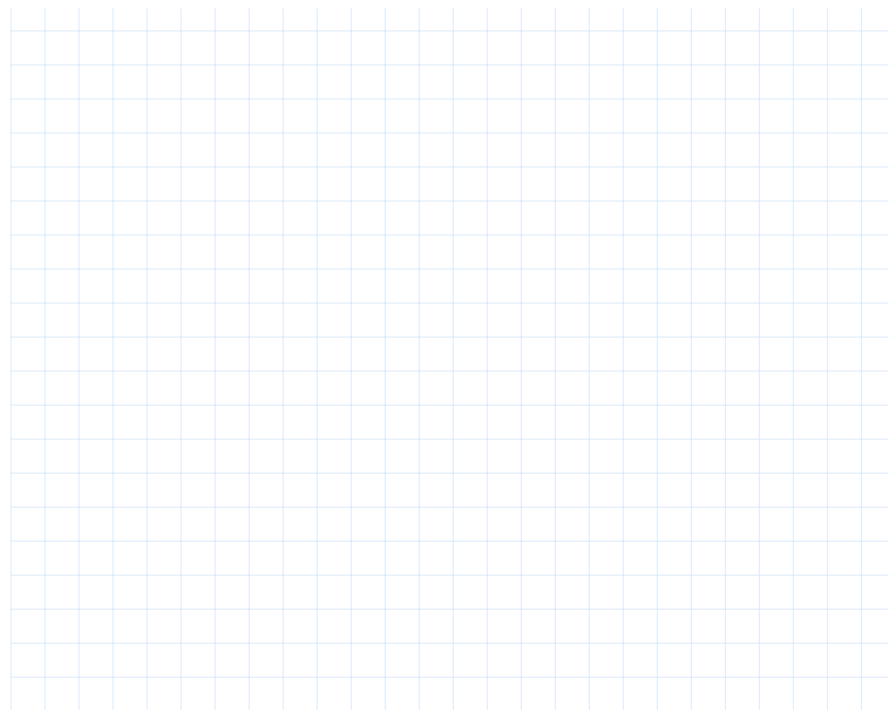
Odd n : symmetry about the origin.

Example. Sketch each graph using $x = -1, 0, 1$ and its overall shape. State the domain and range.

(a) $f(x) = x^4$



(b) $g(x) = x^3$



Leading Term and End Behavior

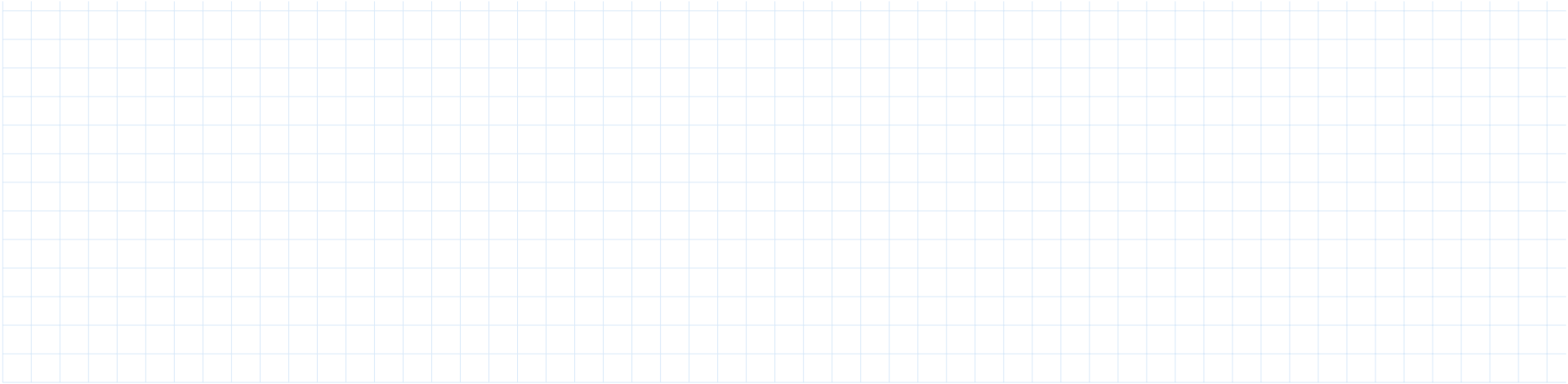
For large $|x|$, a polynomial has the same end behavior as its leading term a_nx^n .

Degree	Leading coefficient	As $x \rightarrow -\infty$	As $x \rightarrow \infty$
Even	$a_n > 0$	$f(x) \rightarrow \infty$	$f(x) \rightarrow \infty$
Even	$a_n < 0$	$f(x) \rightarrow -\infty$	$f(x) \rightarrow -\infty$
Odd	$a_n > 0$	$f(x) \rightarrow -\infty$	$f(x) \rightarrow \infty$
Odd	$a_n < 0$	$f(x) \rightarrow \infty$	$f(x) \rightarrow -\infty$

Example. For $p(x) = -2x^5 + 3x^2 - 7$, identify the leading term and describe both ends. Complete the limits.

$\lim_{x \rightarrow -\infty} p(x) =$

$\lim_{x \rightarrow \infty} p(x) =$

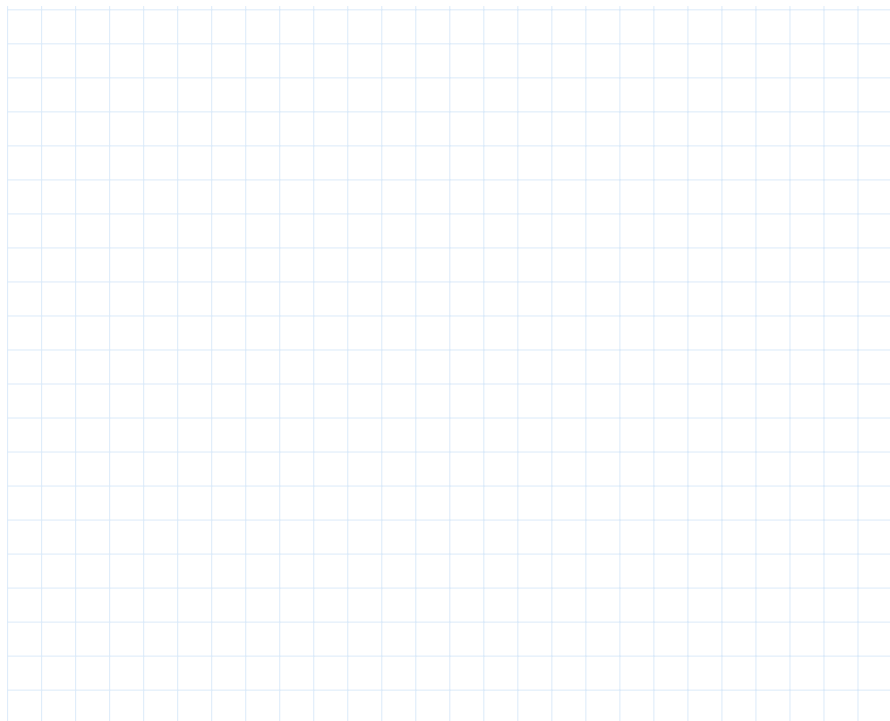


Write the Limits

Example. Use the leading term to determine both limits for each polynomial. Sketch arrows showing its left and right ends.

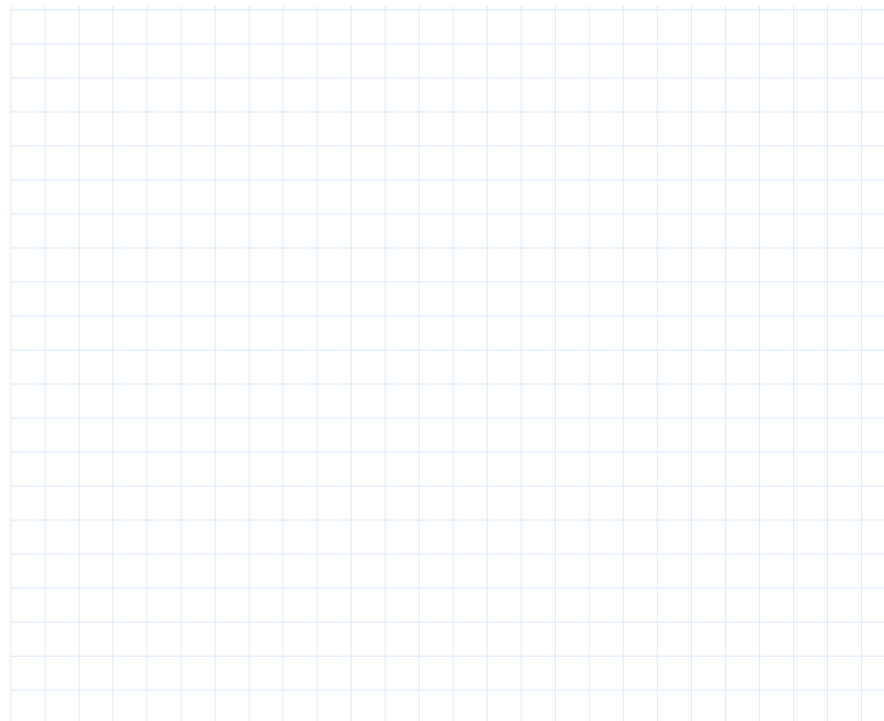
(a) $f(x) = 3x^4 - 5x + 2$

$$\lim_{x \rightarrow -\infty} f(x) = \quad \lim_{x \rightarrow \infty} f(x) =$$



(b) $g(x) = -x^6 + 4x^2$

$$\lim_{x \rightarrow -\infty} g(x) = \quad \lim_{x \rightarrow \infty} g(x) =$$



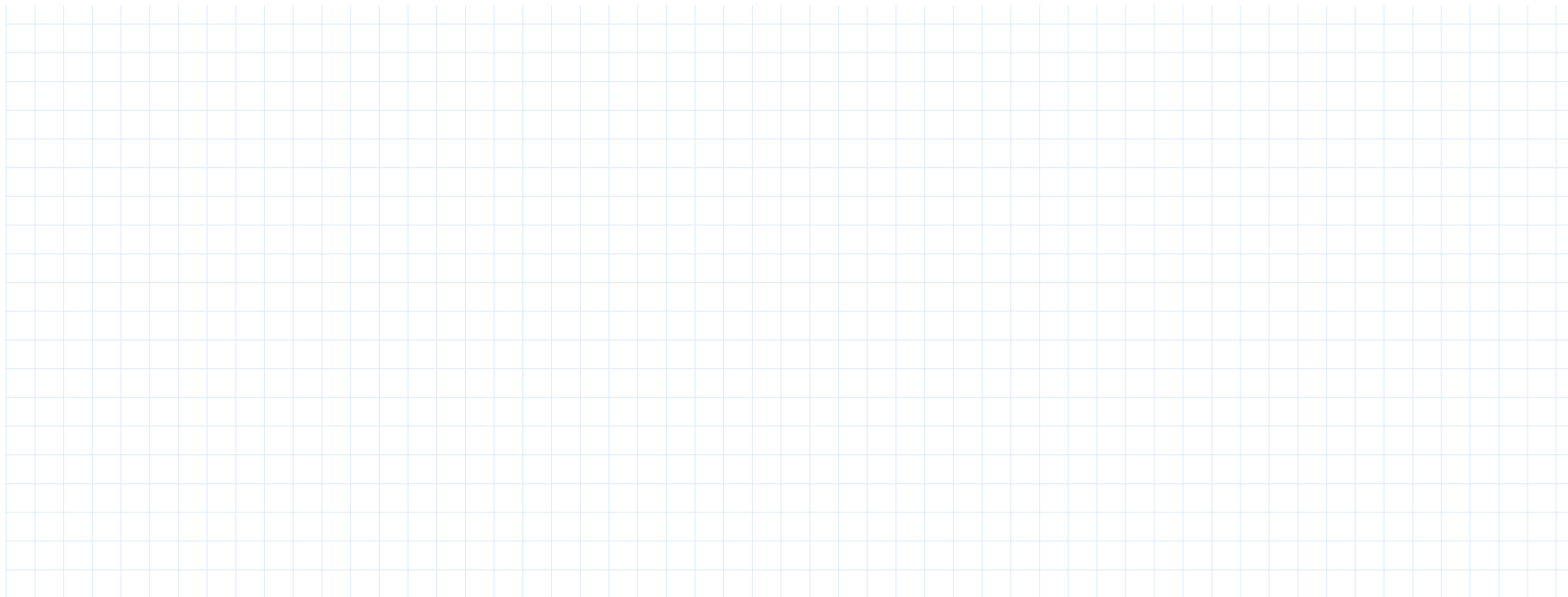
Zeros and Multiplicity

A real number r is a **zero** if $f(r) = 0$; the corresponding intercept is $(r, 0)$.

A factor $(x - r)^m$ gives zero r with **multiplicity** m .

Odd multiplicity: the graph **crosses** the x -axis. Even multiplicity: it **touches and turns**.

Example. Find the zeros of $f(x) = (x + 2)^2(x - 1)^3$. Give each multiplicity and state whether the graph crosses or touches. Find the degree and leading coefficient.



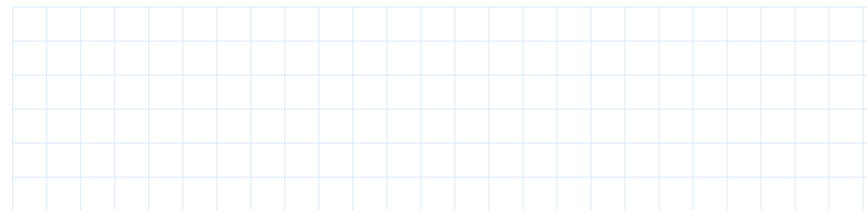
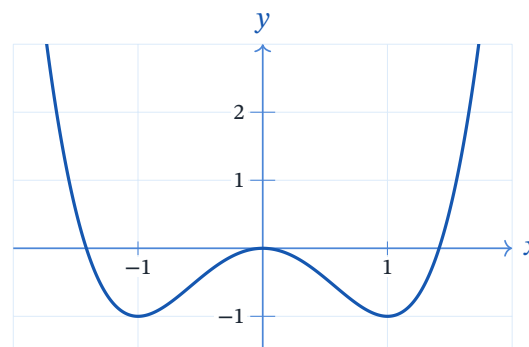
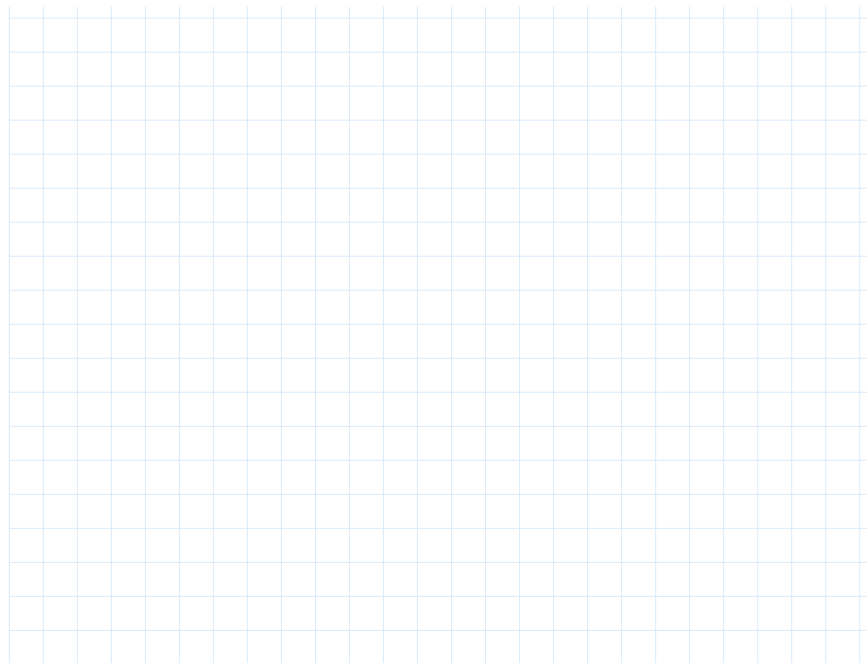
Turning Points

A degree- n polynomial has **at most** $n - 1$ **turning points**.

If a polynomial graph has k turning points, its degree is **at least** $k + 1$.

(a) For $f(x) = -2(x + 1)^2(x - 3)$, find the degree and the maximum possible number of turning points.

(b) Count the turning points in this polynomial graph. What is its minimum possible degree?



Put the Information Together

Example. Let $p(x) = -2(x + 3)(x - 1)^2$.

- (a) Find the degree, leading term, and y -intercept.
- (b) Find the zeros and multiplicities. State crossing or touching at each zero.
- (c) Write both end-behavior limits and give the maximum possible number of turning points.

